

S.No	Description	Theoretical	Practical
1.	R v Pt Pt = 1 : 10 : 10000 (Pt = -9.33 dB) Gt = 31 Gr = 31 λ = 0.03 m σ = 1 m² Pτ = 10⁻¹² W Pt = 8.697 W	$R = \left[\frac{P_t G_r G_r \lambda^2 \sigma}{(4\pi)^3 P_\tau} \right]^{1/4}$ $R = \left[\frac{8.697 \times 30 \times 30 \times (0.03)^2 \times 1}{(4\pi)^3 \times 10^{-12}} \right]^{1/4}$ $= \left[\frac{7.04457}{1.9844 \times 10^{-9}} \right]^{1/4} = 244.1 \text{ m}$	R = 247.1 m
2.	R v Pr Pτ = -15 : 1 : 9 (-108 dB) Gt = 30 Gr = 30 λ = 0.03 m σ = 1 Pt = 1000 Pτ = 1.445 × 10⁻¹¹	$R = \left[\frac{1000 \times 30 \times 30 (0.03)^2 \times 1}{(4\pi)^2 \times 1.44526 \times 10^{-11}} \right]^{1/4}$ $= \left[\frac{810}{3.9594 \times 10^{-8}} \right]^{1/4}$ $= 378.19 \text{ m}$	R = 398.19 m

Compute Parameters like RANGE,
VELOCITY, POWER using RADAR Equation.

Aim:

To compute the maximum detection range and received power of a RADAR system using the RADAR Range equation

APPARATUS REQUIRED:

- * Laptop with i3 processor
- * SCI Lab Software.

CODING:

clc;

clear;

$P_t = \text{~~1000~~};$ 9.33

$G_{\text{dB}} = 31;$

$G = 10^{(G_{\text{dB}}/10)};$

$f = 10e9;$

$c = 3e8;$

$\lambda = c/f;$

$\sigma = 10;$

$L_{\text{dB}} = 3;$

$L = 10^{(L_{\text{dB}}/10)};$

$$S_{min} = 1e-13;$$

$$A_{max} = ((P_t * G^2 * \lambda^2 * \sigma) / ((4 * \pi)^3 * S_{min} * L))^{1/4};$$

disp (" = = = = = ");

disp ("..... RADAR SIMULATION RESULT.....");

disp (" = = = = = ");

disp (" Transmitted Power (w): ");

disp (P_t);

disp (" Antenna Gain (dB): ");

disp (G - dB);

disp (" Frequency (GHz): ");

disp (f / 1e9);

disp (" Wavelength (m): ");

disp (lambda);

disp (" Radar Cross Section (m²): ");

disp (sigma);

~~disp (" System Loss (dB): ");~~

~~disp (L - dB);~~

disp (" Minimum Detectable Signal (w): ");

disp (S_{min});

disp ("Maximum Radar Range (km) : ");

disp (Rmax / 1000);

R = 100:100:10000;

$$P_r = (P_t * G_1^2 * \lambda^2 * \sigma) / ((4 * \pi)^3 * R^4 * L);$$

$$P_r - \text{dBm} = 10 * \log_{10} (P_r / 1e-3);$$

Pt - range = 100:100:10000;

$$R_{\text{max}} - P_t = ((P_t - \text{range} * G_1^2 * \lambda^2 * \sigma) / ((4 * \pi)^3 * \sin^2 L))^{1/4};$$

$$R_{\text{max}} - P_t - \text{km} = R_{\text{max}} - P_t / 1000;$$

G1 - dB - range = 10:1:40;

$$G_1 - \text{range} = 10^{(G_1 - \text{dB} - \text{range} / 10)};$$

$$R_{\text{max}} - G_1 - \text{km} = R_{\text{max}} - G_1 / 1000;$$

clf (1);

subplot (2, 2, 1);

plot (R / 1000, P_r - dBm);

xlabel ("Range (km)");

ylabel ("Received Power (dBm)");

title ("Received Power vs Target Range");

xgrid ();

subplot (2, 2, 2);

plot (Pt - range / 1000, Rmax - Pt - Km);

x label ("Transmitted Power (KW)");

y label ("Maximum Range (Km)");

title ("Maximum Range vs Transmitted power");

x grid();

subplot (2, 2, 3);

plot (G - dB - range, Rmax - G - Km);

x label ("Antenna Gain (dB)");

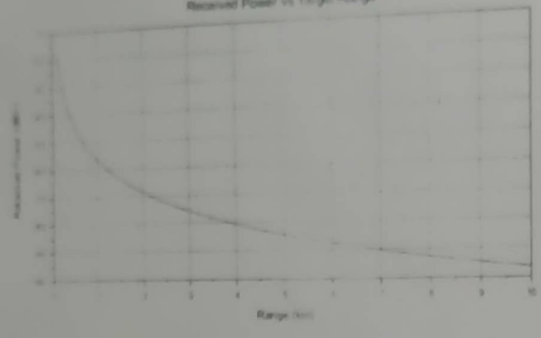
y label ("Maximum Range (Km)");

title ("Maximum Range vs Antenna Gain");

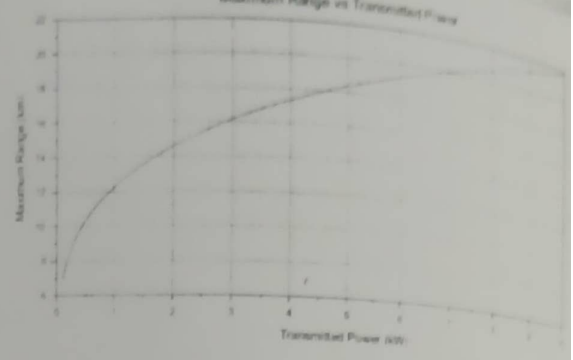
~~x grid();~~

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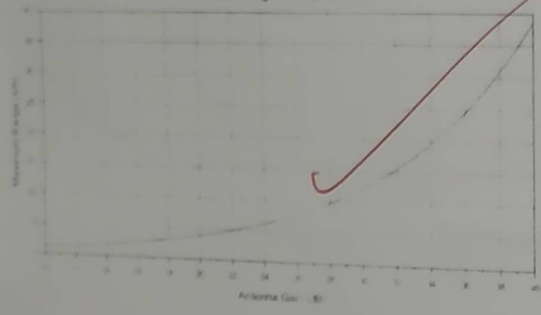
Received Power vs. Target Range



Maximum Range vs. Transmitted Power



Maximum Range vs. Antenna Gain



Scilab program implementation (20)	wave form tabulation (20)	Calculation and Result (20)	Viva voice (20)	Record Result observation (20)	Total mark
20	20	20	16	20	96

Result :

Thus the maximum range of radar system calculated using radar equation is successfully verified in Scilab programming.

Dev